

Class Project

MAE 425 Spacecraft Dynamics and Controls

Group 7

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Satellite Sensors

Thermal Sensors can be used to measure onboard electronics temperature, which will help keep track of the power consumption. They can also be used to place temperature limits on components, such that their temperature does not exceed or fall below the set temperature value.

LiDAR Sensors will be used to maintain a safe distance between InSat-1 and Yellowhammer. They can prove to be very useful in detecting obstacles like space debris so that the satellite can safely maneuver around them and avoid them on time.

Gyroscopes will be used to keep track of the orientation changes of the satellite during maneuvers, ensuring that it is facing the right position. The satellite will be equipped with several gyroscopes to ensure redundancy in case one of them fails. It is critical that the satellite faces the correct direction or it may not be able to receive any last-minute maneuvering instructions updates or send important information to the mission control center.

Accelerometers are installed on InSat-1 to be used for spacecraft navigation. They detect changes in velocity and direction, which are extremely crucial to determining the satellite's position and velocity relative to its surroundings. The necessary forces required to maneuver the satellite are determined by this data.

Magnetometers measure the magnetic field of the planet. Although they are not primarily used for spacecraft navigation, they do provide useful data to determine a satellite's orientation and position relative to the planet it is orbiting.

Optical/Electro-Optical Sensors will be used for taking high-resolution images in a wide range of light spectrum. These images will be taken when InSat-1 maneuvers around Yellowhammer and can be used to analyze the condition of Yellowhammer's critical components.

Hyperspectral Sensors are similar to optical sensors. However, they can capture images in different spectral bands, revealing details that would have otherwise remained hidden. These images will help uncover the condition of Yellowhammer's materials and also its operational status in great detail.

Maneuvers Performed & Mission Timeline

Throughout the mission, twelve maneuvers were performed in total. Of these, the first five were purely prograde burns done at the RAAN point of the initial parking orbit, which were done after an initial waiting period of seven orbits of the initial parking orbit. The sixth was performed at the apogee of the resulting orbit from the first 5 burns, which coincided with the altitude of the target orbit. Here, the burn done was mostly in the normal direction in order to set the inclination to zero, but also had a component in the prograde direction to raise the perigee of the orbit. After this, another three prograde burns occurred, again at apogee, in order to raise the perigee until it had the same altitude as the apogee.

All of these maneuvers were timed in such a way that the chase satellite, upon completing the ninth maneuver to circularize its orbit, will already be exactly 0.25° behind Yellowhammer, so no extra maneuver is needed to get there. After this, one full orbital period went by until the tenth maneuver was performed, which was an entirely retrograde burn with the goal of phasing to the point 0.25° in front of Yellowhammer after one orbital period. At the exact moment that orbital period was waited, and the chase satellite was 0.25° in front of Yellowhammer, the eleventh maneuver was performed, this time entirely prograde with the goal of phasing again, this time to the point 0.10° behind Yellowhammer, again after one orbital period. Once that orbital period went by and the chase satellite was exactly 0.10° behind Yellowhammer, the twelfth and final burn was done to circularize the orbit, which would be an entirely retrograde burn. Detailed information regarding each burn's ΔV cost and timing can be found in Table (1).

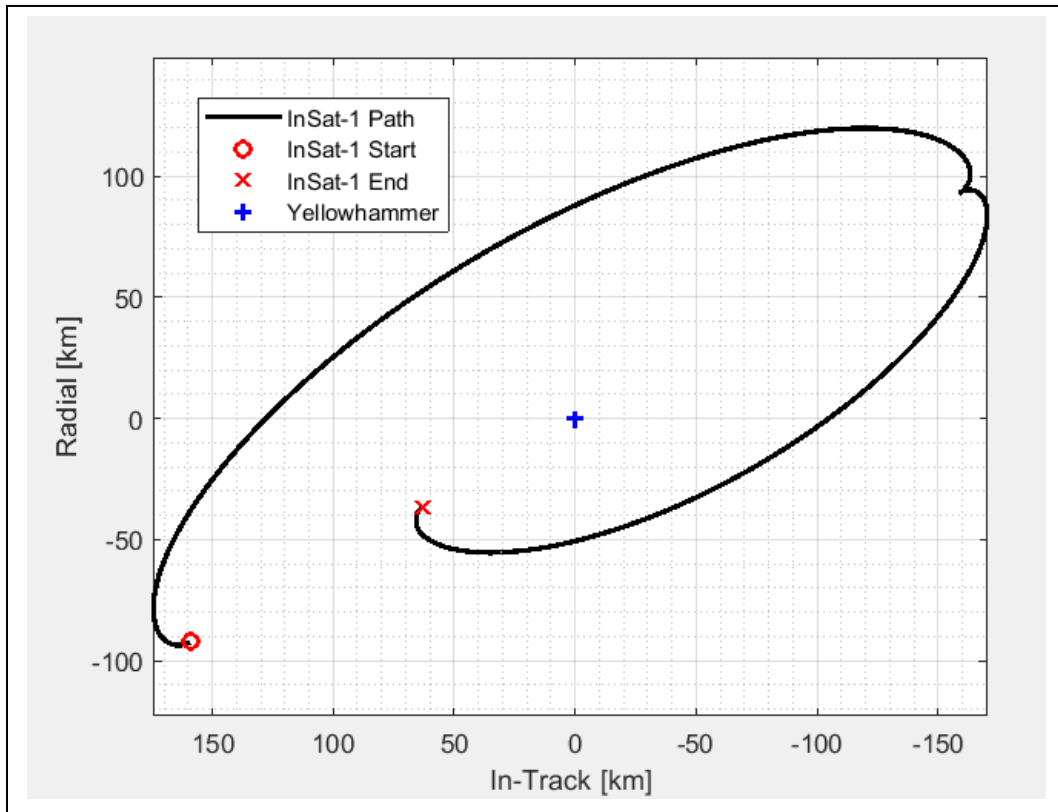
There were many possibilities for paths that could have been taken to complete the mission, this one was chosen specifically with the goal of avoiding doing a burn after initially circularizing into the target orbit to get 0.25° behind Yellowhammer and instead arrive in that exact location. This is why some of the burns for the transfer orbits are of extremely specific ΔV , and why there is an initial waiting period of seven orbits in the initial parking orbit, to make it so the chase arrives in the target orbit at that extremely specific timing. Other than that, it was attempted to further minimize the total ΔV usage, which was accomplished by doing the inclination fix burn at as low a velocity as possible.

	ΔV Cost [m/s]	ΔV Total [m/s]	Interim Time [s]	MJD	Calendar Date
Mission Start	N/A	N/A	0	60341.0000000000	2024-02-01 00:00:00.000 UTC
Burn 1	468.00000000	468.00000000	603144.9938540480	60347.9808448363	2024-02-07 23:32:24.994 UTC
Burn 2	468.00000000	936.00000000	7202.6848822092	60348.0642092446	2024-02-08 01:32:27.679 UTC
Burn 3	468.00000000	1404.00000000	9286.6498504508	60348.1716936179	2024-02-08 04:07:14.329 UTC
Burn 4	468.00000000	1872.00000000	12860.3231435085	60348.3205399506	2024-02-08 07:41:34.652 UTC
Burn 5	457.32872378	2329.32872378	20017.1232942536	60348.5522196183	2024-02-08 13:15:11.775 UTC
Burn 6	349.88554206	2679.21426584	19194.5401212333	60348.7743786475	2024-02-08 18:35:06.315 UTC
Burn 7	468.00000000	3147.21426584	40681.7214571019	60349.2452319051	2024-02-09 05:53:08.037 UTC
Burn 8	468.00000000	3615.21426584	49564.8489249462	60349.8188991381	2024-02-09 19:39:12.886 UTC
Burn 9	317.68598414	3932.90024998	65878.2952590382	60350.5813794073	2024-02-10 13:57:11.181 UTC
Burn 10	1.42543659	3934.32568657	86163.5705505783	60351.5786429553	2024-02-11 13:53:14.751 UTC
Burn 11	2.42088849	3936.74657506	86043.8989248135	60352.5745214151	2024-02-12 13:47:18.650 UTC
Burn 12	0.99545190	3937.74202696	86247.3406886135	60353.5727545249	2024-02-13 13:44:45.991 UTC
Mission End	N/A	N/A	0	60353.5727545249	2024-02-13 13:44:45.991 UTC

Table (1): ΔV and Timing Information for Each Maneuver

As seen in the table, the **total ΔV** cost for the entire mission is **3937.74 m/s**, and the ending completion date and time is exactly **February 13th, 2024 at 08:44:46 AM EST**.

Relative Motion of InSat-1 around Yellowhammer.



Ground Trace Plot of InSat-1 and Yellowhammer.

